

The speed of sound in a two-dimensional hard-disk gas: a divider-mode measurement against the equation of state

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Abstract

We measure the adiabatic speed of sound of a hard-disk fluid from the eigenfrequency of a massive divider separating two compartments, the method of Román et al. (2002), over 35 packing fractions from $\eta = 0.0065$ to 0.76 and nine divider masses per density. The measurement is compared against the Kolafa–Rottner equation of state through the adiabatic mapping $c_s^2 = (k_B T/m)[Z + \eta Z' + Z^2]$ with no fitted parameter. For $\eta \leq 0.4$ the agreement is +1.0 % on average and never worse than 1.7 % at $N = 100$; extrapolating to infinite system size along a ladder of $N = 100$ –2500 brings it to $\lesssim 1\%$. We give the finite-size law, the estimator and its sensitivity, and a divider linewidth that recovers the Enskog transport coefficient of the same gas.

1 Introduction

Why the divider mode, why hard disks, what is being tested against what. State plainly that the EOS is an input, not a fit, so agreement is a test and disagreement is a measurement.

2 Model and geometry

Hard disks of diameter $\sigma = 2r$ and mass m in a box $2L_0 \times H$, split by a divider of mass M and thickness t at the centre; N_s disks each side; $k_B T = m = \sigma = 1$; event-driven dynamics with the scientific backend (`--edmd-acc=0`). The divider is held for an equilibration period and then released; its position is recorded.

Effective length. The centres of the disks in one compartment span r to $L_0 - t/2 - r$, so

$$L_{\text{eff}} = L_0 - 2r - t/2, \quad t = 0.05 \sigma. \quad (1)$$

Carrying t lowers every c_s by $(t/2)/(L_0 - 2r)$: 0.07 % at $\eta = 0.10$ and 0.50 % at $\eta = 0.65$.

3 The estimator

$$P(f) = \left| \sum_j (x_j - \bar{x}) e^{-2\pi i f t_j} \right|^2, \quad \nu = \arg \max_{f \geq \nu_{\text{pred}}/X} P(f), \quad X = 2.5, \quad (2)$$

per trajectory, then the mean over seeds, then c_s as the through-origin slope of $\bar{\nu}_M$ against $x_M = K(\alpha)/(2\pi L_{\text{eff}})$ with $\cot K = \alpha K$, $\alpha = M/2N_s m$. The one-sided edge exists because the divider carries a second, slow mode; the choice $X = 2.5$ is flat over $X \in [2.08, 6.25]$ at 200 periods.

Why the largest bin and not a resonance fit. Both put the nine masses on one line to 0.27%, but the residual of the fit *leans* with mass, $-0.286 \pm 0.020\%$ per e-fold in α (14.4σ), while the largest bin is flat, $-0.030 \pm 0.029\%$ (1.0σ). The fit is the more precise estimator and the more biased one.

4 Results

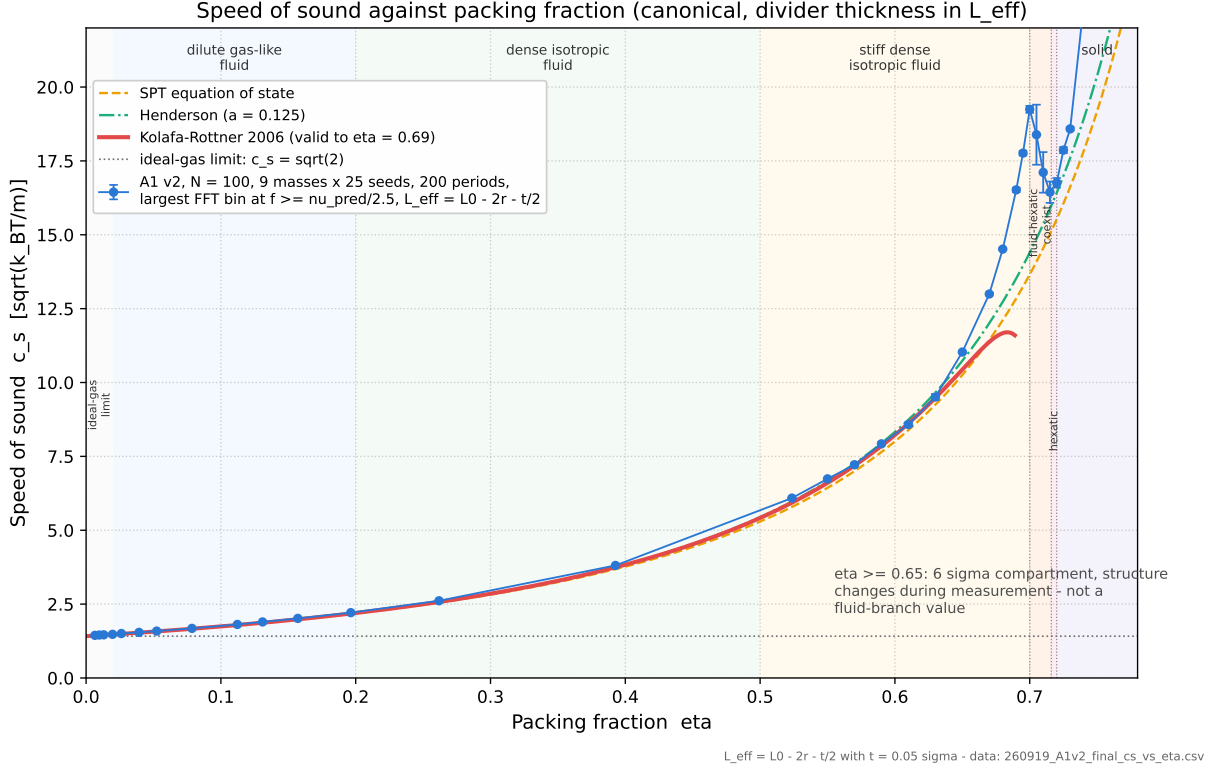


Figure 1: $c_s(\eta)$ at $N = 100$ against three equations of state. Shading marks the approximate regimes; only 0.700/0.716/0.720 are physical transitions. Above $\eta = 0.65$ the compartment is 6σ wide and the structure changes during the measurement, so those points are not fluid-branch values.

Over $\eta \leq 0.4$ the measurement sits $+1.04\%$ above Kolafa–Rottner on average, with a maximum deviation of 1.68%. 7873 of 7875 trajectories were used; two were discarded by the health contract.

5 Finite size

Along $N = 100, 400, 900, 1600$ (and 2500 where available) at fixed η , $c_s(N)$ is fitted with $c_\infty + b/\sqrt{N}$; the two other forms (b/N and both together) bracket the systematic. At $\eta = 0.02$, $c_\infty = 1.4614 \pm 0.0102$ against 1.4725 from the EOS (-1.1σ); at $\eta = 0.05$, 1.5699 ± 0.0029 against 1.5666 ($+1.1\sigma$).

What the offset is not. The wall pressure of the same finite box is 7–20% above the bulk EOS and rising with η , which would imply a $+5$ to $+16\%$ excess in c_s — five times what is observed. The wall value is a surface contact value; the mode samples the bulk. The finite-size law above is the measurement, not the wall pressure.

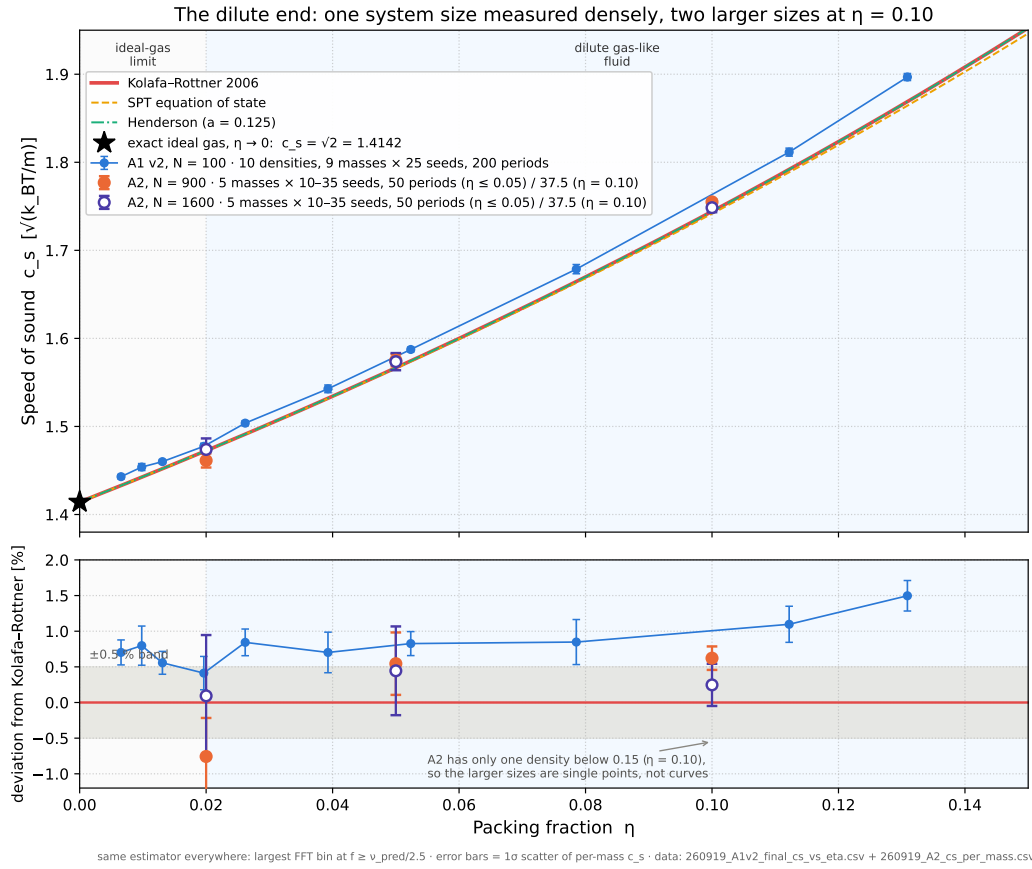


Figure 2: The dilute end. The $N = 100$ curve sits above the EOS by 0.5–1.5%; the larger systems agree with it within their errors, so the offset is finite size.

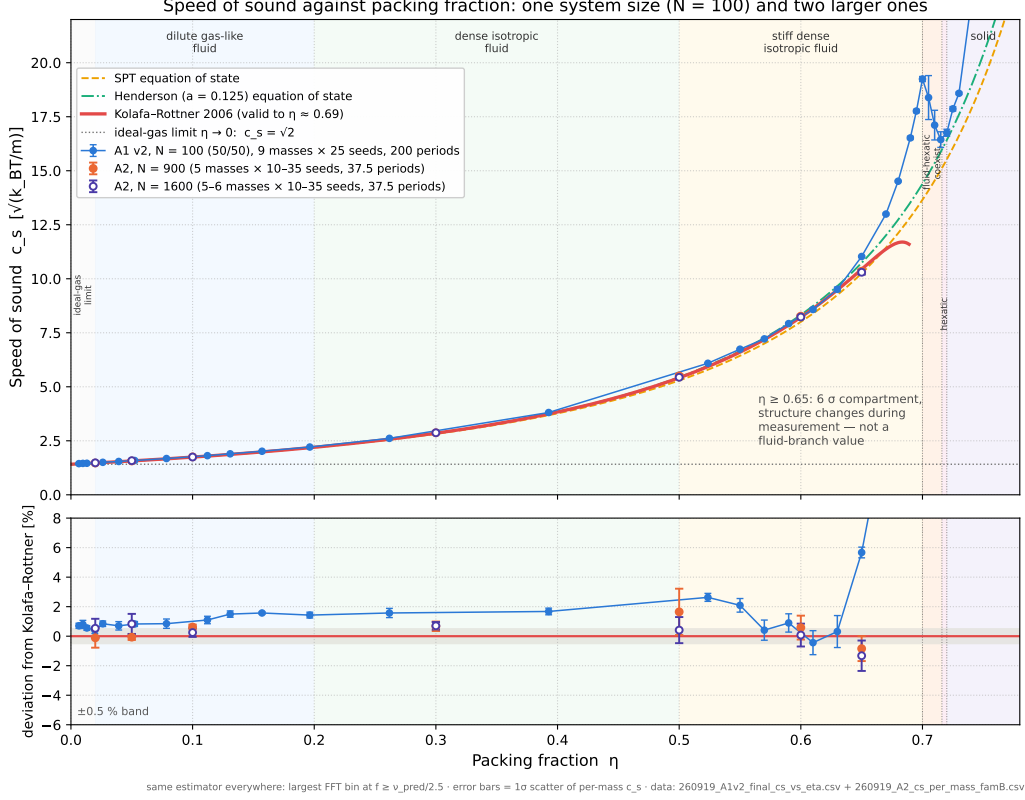


Figure 3: $N = 100$ against $N = 900$ and 1600 .

6 The divider linewidth gives a transport coefficient

Linearising the piston equation of Malek Mansour et al. gives a damped oscillator with $Q = 1/\mu$, $\mu = \Gamma L_y \sqrt{2/(\hat{M}N)}$, $\hat{M} = M + mN/3$. The measured FWHM of the seed-averaged spectrum, with the bin width removed in quadrature, gives $\Gamma = 0.314 \pm 0.038$ at $\eta = 0.112$, against the Enskog value 0.331 — consistent within its 12% error. The $M = 50$ row ($\alpha = 0.5$) fails the $K \ll 1$ assumption and is excluded; the $M = 500$ row is 0.59 of prediction and is not explained.

7 Methods

Health contract, seeding, sampling, record length, discard rule.

8 Reproducibility

Every campaign directory carries `00_COMMAND.md` with the exact command and the estimator geometry line. The geometry constant and the estimator share one definition (`tests_20260913.WALL.T`, `l_eff`), and `assert_wall_thickness` refuses to analyse data whose recorded thickness disagrees.

Open

- $\eta > 0.69$: no fluid reference. Liu (2021) global EOS with hexatic and solid branches.
- $\eta = 0.65$ at larger N , and an $N \approx 1000$ sweep, as the final Paper 1 runs.

- The $M = 500$ linewidth row.